

Delays in Computer Network

Transmission Time • Propagation Time • Throughput • Link Utilization

Delays in Computer Networks

Data experiences four types of delays in networks:

1. PROPAGATION DELAY

$$P_d = d / V$$

Time for 1 bit to traverse link

2. TRANSMISSION DELAY

$$T_d = \text{Data Size} / \text{Bandwidth}$$

Time to push bits onto wire

3. QUEUING DELAY

Packet waits in router buffer

Variable - depends on congestion

4. PROCESSING DELAY

Router processes header

Error checking, routing decisions

Total Delay (Nodal Delay)

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

Transmission Time/Delay (T.T)

Transmission Time is the time taken to push all the packet bits onto the transmission medium.

It is a property of the **computer/sender** - depends on data size and link bandwidth.

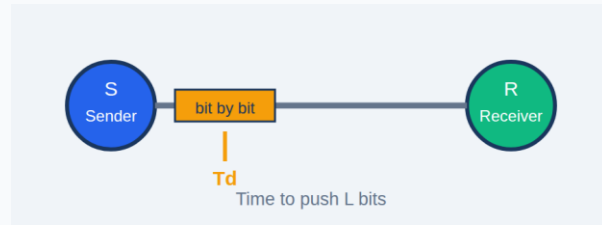
Formula

$$T.T = \text{Data Size} / \text{Bandwidth}$$

(bits / bits per second = seconds)

Note: Maximum link capacity is known as **Bandwidth**

Transmission Time Visualization



Unit Conversions:

Kilo = 10^3 , Mega = 10^6 , Giga = 10^9

milli = 10^{-3} , micro = 10^{-6} , nano = 10^{-9}

Example: Transmission Time Calculation

Question 1: Calculate the Transmission Time

Given: Data = 2 Kbits, Bandwidth = 10 Mbps

Solution:

$$T.T = \text{Data Size} / \text{Bandwidth}$$

$$T.T = 2 \times 10^3 \text{ bits} / (10 \times 10^6 \text{ bits/sec})$$

$$T.T = 2 \times 10^3 / 10^7 = 2 \times 10^{-4} \text{ sec}$$

$$\mathbf{T.T = 200 \text{ }\mu\text{sec (microseconds)}}$$

Key Insight: T.T depends on how fast sender pushes bits onto link (bandwidth) and data size.

Propagation Time/Delay (P.T)

Propagation Time is the time taken by 1 bit to traverse the link from source to destination.

It is a characteristic of the **link/medium**.

Formula

$$P.T = d / v$$

(meters / meters per second = seconds)

Important: P.T does NOT depend on data size!

Propagation Delay Visualization



Example: $d = 2 \text{ km}$, $V = 2 \times 10^8 \text{ m/s}$

$$P.T = 2 \times 10^3 / 2 \times 10^8 = 10^{-5} \text{ sec} = 10 \text{ } \mu\text{sec}$$

Typical Velocity: $V \approx 2 \times 10^8 \text{ m/s}$ (2/3 speed of light)

Round Trip Time (R.T.T)

Round Trip Time is the time for a signal to travel from source to destination and back.

Formula

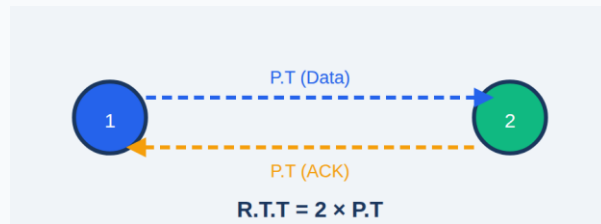
$$R.T.T = 2 \times P.T$$

R.T.T is a **characteristic of the link**.

Total Time for Data Transfer:

$$\text{Total Time} = T.T + 2 \times P.T$$

Round Trip Time Diagram



Note: ACK size $\ll$ Data size, so T.Tack is negligible

Throughput

Throughput is the rate at which data is actually transmitted by the user over the network.

Bandwidth $\neq$ Throughput

Bandwidth is max capacity; Throughput is actual usage

Formula

$$\text{Throughput} = \text{Data Size} / (\text{T.T} + 2 \times \text{P.T})$$

Throughput changes due to propagation delay and ACK waiting time.

Throughput vs Bandwidth

Bandwidth (Max Capacity)

Actual Throughput

Unused

Alternative Forms:

$$\text{Throughput} = (\text{Data Size} / \text{B.W}) / \text{Total Time}$$

$$\text{L.U} = \text{Throughput} / \text{B.W}$$

Example: Throughput Calculation

Question: Calculate the Throughput

Given: Data = 100 bits, B.W = 10 Mbps, $d = 2.3$ km, $V = 2.3 \times 10^8$ m/sec

Solution:

Step 1: Calculate T.T

$$T.T = 100 \text{ bits} / 10^7 \text{ bits/sec}$$

$$T.T = 10^{-5} \text{ sec} = 10 \text{ } \mu\text{sec}$$

Step 2: Calculate P.T

$$P.T = 2.3 \times 10^3 / 2.3 \times 10^8$$

$$P.T = 10^{-5} \text{ sec} = 10 \text{ } \mu\text{sec}$$

$$\text{Step 3: Throughput} = 100 \text{ bits} / (10 + 2 \times 10) \text{ } \mu\text{sec} = 100 / 30 \text{ } \mu\text{sec}$$

$$\text{Throughput} = 3.33 \text{ Mbps (33.3\% of Bandwidth)}$$

Link Utilization (L.U)

Link Utilization measures sender's efficiency in using the transmission link.

Formula 1 (Time-based)

$$L.U = T.T / (T.T + 2 \times P.T)$$

Formula 2 (Rate-based)

$$L.U = \text{Throughput} / \text{Bandwidth}$$

$$\%L.U = L.U \times 100$$

Example:

Throughput = 3.3 Mbps, B.W = 10 Mbps

$$\%L.U = (3.3/10) \times 100 = 33.3\%$$

Key Insight: Higher L.U = better efficiency. Low L.U means link is underutilized.

When L.U = 50%: $T.T = 2 \times P.T$

Example: Number of Bits in R.T.T

Q7: Calculate number of bits in R.T.T

Given: B.W = 100 Mbps, R.T.T = 50 μ sec

Solution:

Step 1: Bits in 1 second = 100 Mbps = 10^8 bits

Step 2: Bits in 50 μ sec = $50 \times 10^{-6} \times 10^8$ bits

Number of bits in R.T.T = $BW \times RTT = 5000$ bits

Concept: This represents bits "in flight" on the link - important for sliding window protocols!

Example: Data Units in R.T.T

Q8: Calculate number of data units in R.T.T

Given: B.W = 10 Mbps, R.T.T = 40 μ sec, Data Size = 20 bits

Solution:

Step 1: 1 sec $\rightarrow 10^7$ bits

Step 2: 40 μ sec $\rightarrow 40 \times 10^{-6} \times 10^7 = 400$ bits

Step 3: No. of Data Units = Total bits / Data Size
 $= 400 / 20 = 20$

Number of Data Units in R.T.T = 20

Application: Determines window size in sliding window protocols.

1-Bit Delay Concept

1-bit delay is the time required to place 1 bit on the transmission medium.

Formula

$$\text{1-bit delay} = 1 / \text{Bandwidth}$$

Q5: B.W = 1000 Mbps (1 Gbps)

1 sec $\rightarrow 10^9$ bits

1-bit delay = 10^{-9} sec = **1 nanosec**

1-bit delay in meters of cable

$$\text{Distance} = V \times \text{1-bit delay}$$

Q6: B.W=100Mbps, V=2×10⁸ m/s

1-bit delay = 10^{-8} sec

Distance = $2 \times 10^8 \times 10^{-8} = \text{2 meters}$

Insight: At 100 Mbps, 1 bit occupies 2m of cable!

Special Case: When L.U = 50%

Q3: Prove that when L.U = 50%, T.T = 2 × P.T

Derivation:

$$L.U = T.T / (T.T + 2 \times P.T)$$

$$0.5 = T.T / (T.T + 2 \times P.T)$$

$$0.5 \times T.T + P.T = T.T \rightarrow P.T = 0.5 \times T.T$$

$$\therefore T.T = 2 \times P.T \text{ (when L.U = 50\%)}$$

Q4: Find Data Size when L.U=50%, d=2km, V=2×10⁸m/s, B.W=10Mbps

$$\text{Data Size/B.W} = 2 \times d/V \rightarrow x/10^7 = 2 \times 2 \times 10^3 / 2 \times 10^8 \rightarrow x = 200 \text{ bits}$$

Queuing Delay

Queuing Delay is the time packet waits in router buffer before processing.

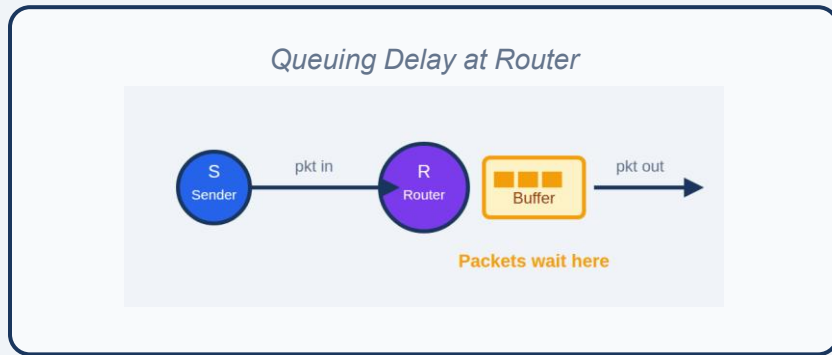
Characteristics:

Variable delay - depends on network congestion

Can be zero if queue is empty

Can be very large if congested

Warning: If queue is full, packets are dropped!



Processing Delay

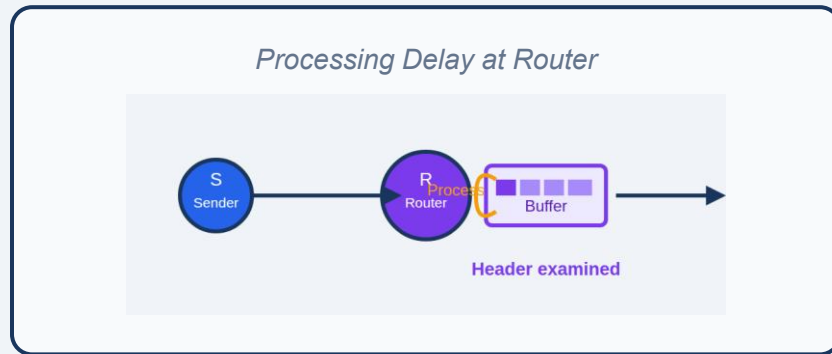
Processing Delay is the time router takes to process the packet header.

Processing includes:

- Examining packet header
- Checking for bit-level errors
- Determining output link
- Routing table lookup

Characteristics:

- Typically very small (microseconds)
- Usually negligible compared to other delays



Buffer and Timer for Retransmission

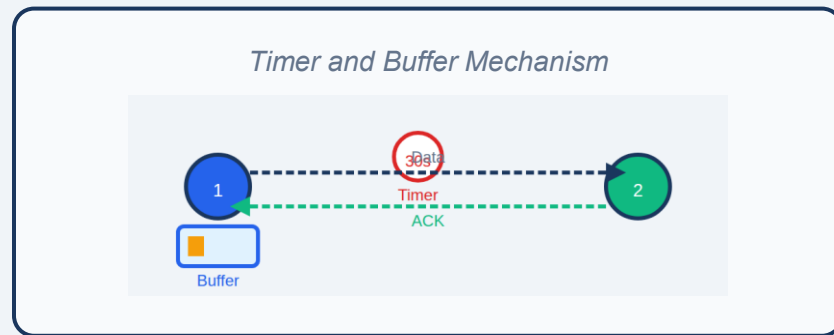
Reliable data transfer requires **Timer** and **Buffer**.

Why Timer?

When timer expires without ACK → Retransmit
Waiting time is finite (cannot wait forever)

Why Buffer?

Stores data until ACK is received
Required to retransmit if data is lost



If Data Lost: Timer expires → Retransmit → Restart timer

Summary: Key Formulas

TRANSMISSION TIME

$$T.T = \text{Data Size} / \text{Bandwidth}$$

Property of sender/computer

PROPAGATION TIME

$$P.T = \text{Distance} / \text{Velocity}$$

Property of link ($V \approx 2 \times 10^8$ m/s)

ROUND TRIP TIME

$$R.T.T = 2 \times P.T$$

Total Time = $T.T + 2 \times P.T$

1-BIT DELAY

$$\text{1-bit delay} = 1 / B.W$$

In meters: $V \times (\text{1-bit delay})$

THROUGHPUT

$$\text{Throughput} = \text{Data} / (T.T + 2 \times P.T)$$

Actual data rate ($\leq$ Bandwidth)

LINK UTILIZATION

$$L.U = T.T / (T.T + 2 \times P.T)$$

$$L.U = \text{Throughput} / \text{Bandwidth}$$

When $L.U = 50\%$: $T.T = 2 \times P.T$

FOUR NETWORK DELAYS

1. Propagation ($P_d = d/V$)
2. Transmission ($T_d = L/B$)
3. Queuing (variable)
4. Processing (negligible)